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Application for Testing and Measuring multi-RAT

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Indhold

1 Introduction

In many transportation systems, it is logical to have multiple roads that connect two neighboring cities, ensuring that the connection between them is less likely to be disrupted by accident or congestion. A complementary principle can be applied to when sending an important letter. By simultaneously delivering duplicate letters using different roads, it increases the chance that at least one copy of the letter makes it to the intended location. Furthermore, once one letter arrives, any arriving copies can simply be disregarded.

The modern communication system generally uses singular wireless technologies such as WiFi, LTE, and 5G to provide connectivity. This is equivalent to connecting the two cities with only a single road. This is unstable due to interference, congestion, mobility, and varying channel conditions. All of this can introduce increased latency, packet loss, jitter, and reduced throughput, which decreases the quality of service (QoS)

Multiple Radio Access Technology, Multi RAT, is the system that implements multiple roads, ensuring reliable and timely data delivery over wireless networks. It uses the previously mentioned wireless technologies to simultaneously deliver duplicate packets using different radio access technology. The receiver processes the first successfully received packet and discards duplicates, this is called 1+1 Protection.

2 Theory

Each radio access technology that is compatible with Multi RAT has its own characteristics in terms of coverage, bandwidth, latency, and reliability. Using Multi RAT allows these technologies to combine their strengths and cover their weaknesses.

This project uses Wi-Fi and LoRaWAN; LoRaWAN forms a direct long range connection from the sender and receiver. In contrast Wi-Fi uses various access points in between the sender and receiver. Although, Wi-Fi has a much higher capacity than the LoRaWAN, but unlike LoRaWAN, it accumulates additional latency with every network hop.

This, theoretically, means that in shorter distances that consist of only a few network hops, Wi-Fi will be much more efficient and practical than LoRaWAN. However, once the number of network hops increases, a direct connection like LoRaWAN will eventually become more efficient and practical.

2.1 Quality of Service

Figurer

Tabeller

Litteratur